**SMARTBRIDGE**
**SKILLWALLET**
*A SmartBridge Product*

# Scalability & Future Plan

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| **Date** | 12 July 2026 |
| **Team ID** | **SWTID-2026-6128** |
| **Project Name** | Emotion Detection & Learning Support Engine |
| **Maximum Marks** | 1 Mark |

## Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future.

### Current System Limitations:

| S.No | Limitation | Impact | Priority to Address (High/Medium/Low) |
|---|---|---|---|
| 1 | CSV-based storage doesn't scale well beyond a single-machine prototype. | Concurrent writes from many students could corrupt or slow down the log file. | Medium |
| 2 | The Gemini API is not yet wired in; guidance is currently static per emotion via the mock handler. | Responses don't yet reflect the specific nuance of each student's actual text. | High |
| 3 | No user authentication or per-student history exists. | Can't track an individual student's progress across multiple sessions. | Low |

### Scalability Plan:

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User Load | Single local Streamlit session, informally tested | Deploy to Streamlit Community Cloud or a small cloud VM with session isolation for classroom-scale concurrent use |
| 2 | Data Storage | Flat CSV file (data/interactions_log.csv) | Migrate to a lightweight database (e.g., SQLite, then PostgreSQL) as interaction volume grows |

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|---|---|---|---|
| 3 | Performance | Both BiLSTM and BERT models loaded fully at Streamlit startup | Cache models with st.cache_resource and consider a lighter model for real time inference |
| 4 | Security | No authentication; Gemini key stored only in a local .env | Add basic student login and move secrets to a proper secrets manager for hosted deployment |

## Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|
| Phase 1 | Consolidate mock model conflicts and unify the logger schema | Aug 2026 | Reliable, consistent emotion detection and clean logging data |
| Phase 2 | Wire in the real Gemini API and replace the BERT mock with the trained model | Sep 2026 | Genuinely personalized, input specific guidance |
| Phase 3 | Migrate storage to SQLite/PostgreSQL and add basic student login | Oct 2026 | Supports per-student history and safer concurrent access |
| Phase 4 | Deploy to a hosted environment with classroom-scale load testing | Nov 2026 | Validated to support a full class using the tool concurrently |